Context Aware BGM

Automated background music generation for film scenes using NLP-driven cues

Our Intent...

Our project focuses on **context-aware background music descriptor generation from movie scripts**.

In films, background music plays a crucial role in shaping how a scene is perceived — it conveys emotion, builds tension, and enhances storytelling. However, most existing automated systems rely only on **basic emotion or sentiment classification**, which is not sufficient.

Instead of directly generating music, our system outputs **structured music descriptors** such as tempo, intensity, and texture, which can later be used for music generation or recommendation.

Overall, our goal is to bridge the gap between **story understanding and music design**, making automated film scoring more context-aware and meaningful.

Motivation

Filmmakers need scalable ways to generate mood-consistent music across many scenes. Manual scoring is time-consuming and costly, and generic libraries often lack scene specificity.

By leveraging NLP to interpret scene intent, the system enables **emotional alignment** between audio and narrative while reducing production time.

Joker: “You have nothing to threaten me with… nothing to do with all your strength.”

Batman: “Where is she?”

Joker (laughing): “You think you can control things? You think you can make the rules?”

Batman: “Tell me where they are!”

Joker: “You complete me.”

Emotion: **Psychological tension, chaos, dominance**
Not fast-paced → **slow, intense, unsettling**
Music needed: **low tempo, eerie, minimal, dissonant**

Joker (shouting while riding): “Come on! Hit me! Hit me!”

Joker (laughing wildly): “I want you to do it!”

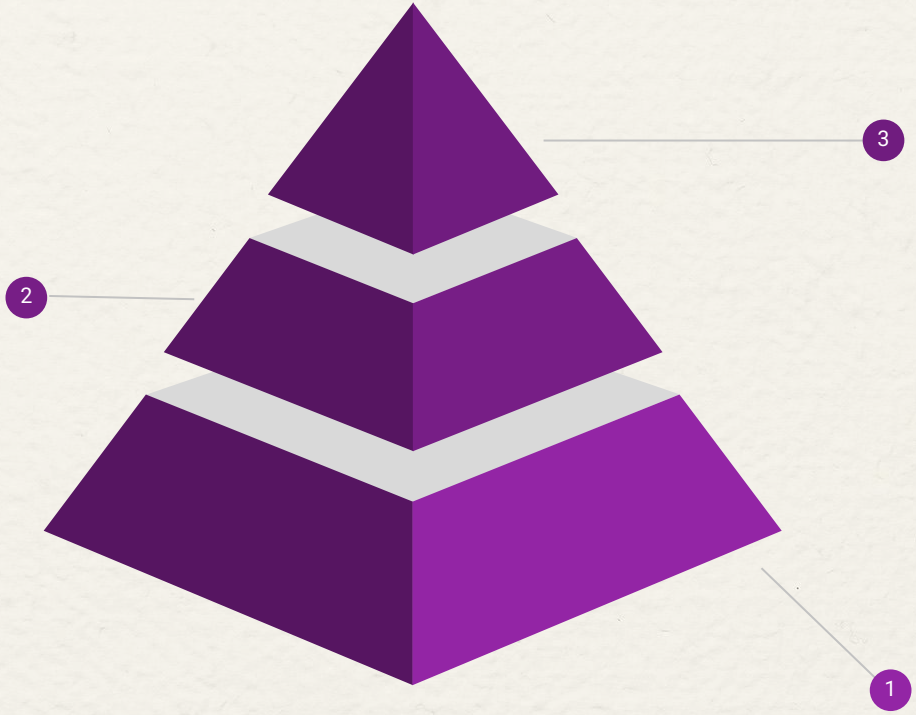
(Batman speeds forward, engines roaring, no reply — pure action focus)

Emotion: **Adrenaline, danger, chaos**
Fast-paced → **high energy, continuous motion**
Music needed: **fast tempo, heavy percussion, intense build up**

Multilevel Annotation

2. Context-Aware Features (Group 2)

- Narrative Arc Position (Setup → Climax → Resolution)
- Tension Level & Arousal Level
- Emotional Shift Detection
- Foreshadowing & Transitions



3. Music Descriptor Labels (Output Targets)

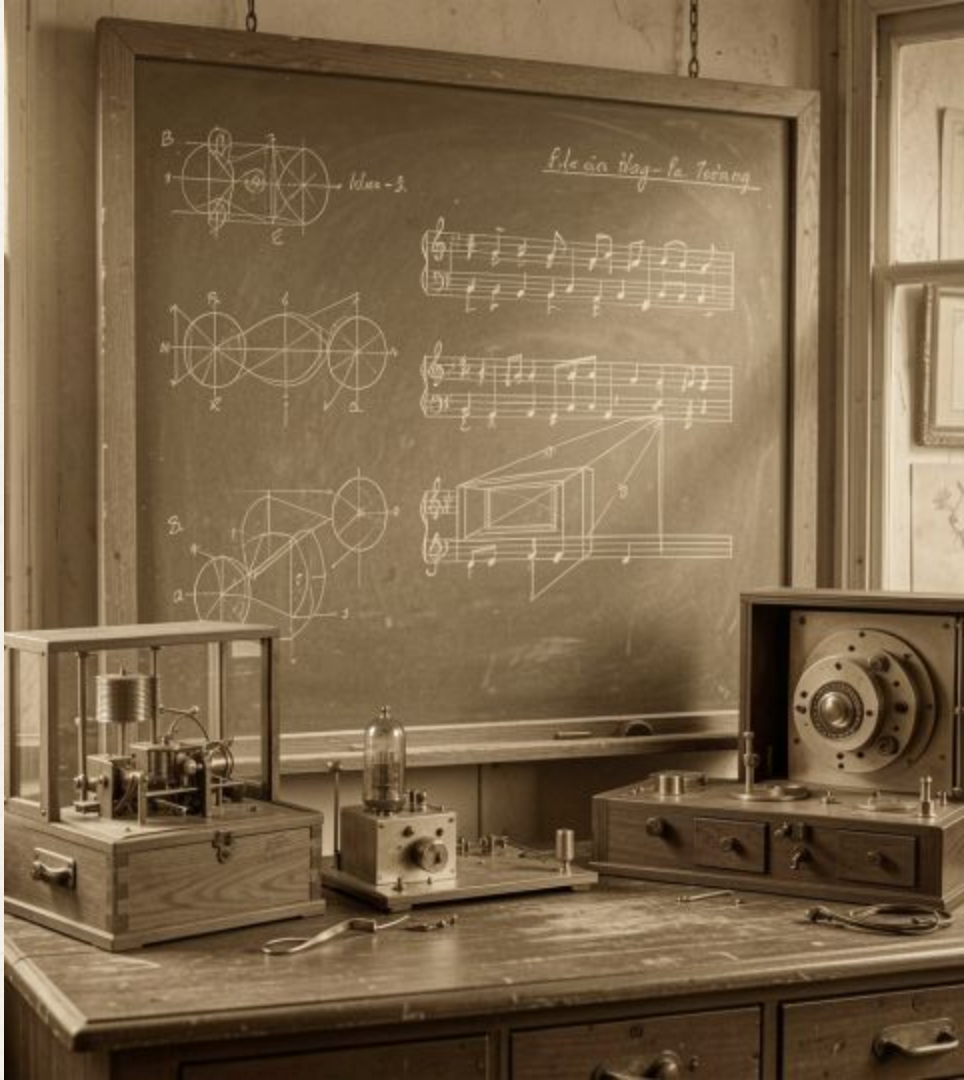
- Tempo (BPM), Musical Valence
- Tonality & Harmonic Style
- Orchestration (multi-label instruments)
- Rhythm Style & Texture

1. Scene-Level Features (Group 1)

- Emotional Valence (4 classes)
- Pacing Intensity (1–10)
- Conflict Nature
- Action Intensity
- Acoustic Space & Reality Layer
- Emotion Tags (multi-label)

01

Technical Approach



Scene Perception

- Extracts scene-level embedding (256-d)
- Captures:
  - Emotional Valence
  - Conflict Nature
  - Interaction Tone
  - Acoustic Space & Reality Layer
- Models intensity:
  - Pacing, Action, Tension, Arousal
- Multi-label emotions (Anger, Joy, Fear, etc.)

Output: Rich scene representation vecter

Narrative Modeling

- Takes sequence of scene embeddings
- Captures:
  - Narrative progression
  - Emotional transitions across scenes
  - Tension build-up / release
- Computes:
  - Scene-to-scene changes (delta features)
  - Temporal dependencies

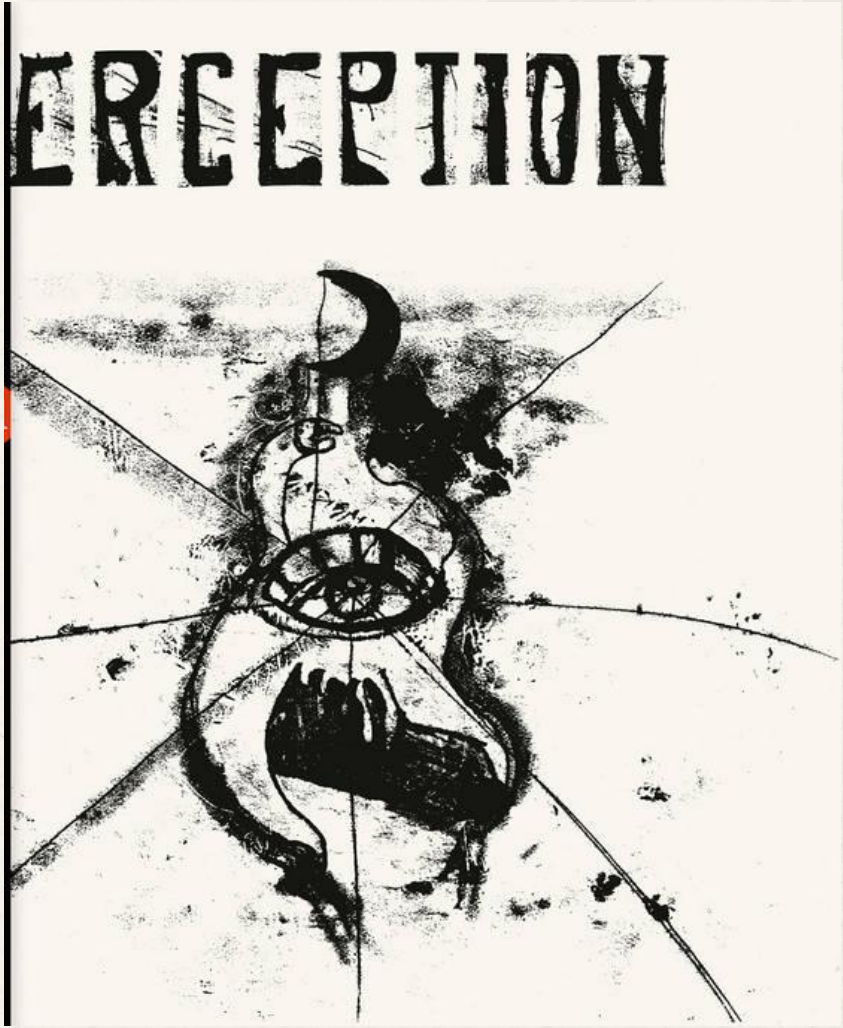
Output: Context-aware scene representation

Music Descriptor Generation

- Maps context-aware features → music space
- Predicts:
  - Tempo (BPM)
  - Tonality (Major/Minor)
  - Musical Valence
  - Orchestration (instruments)
  - Rhythm & Texture

Output: Structured music descriptors

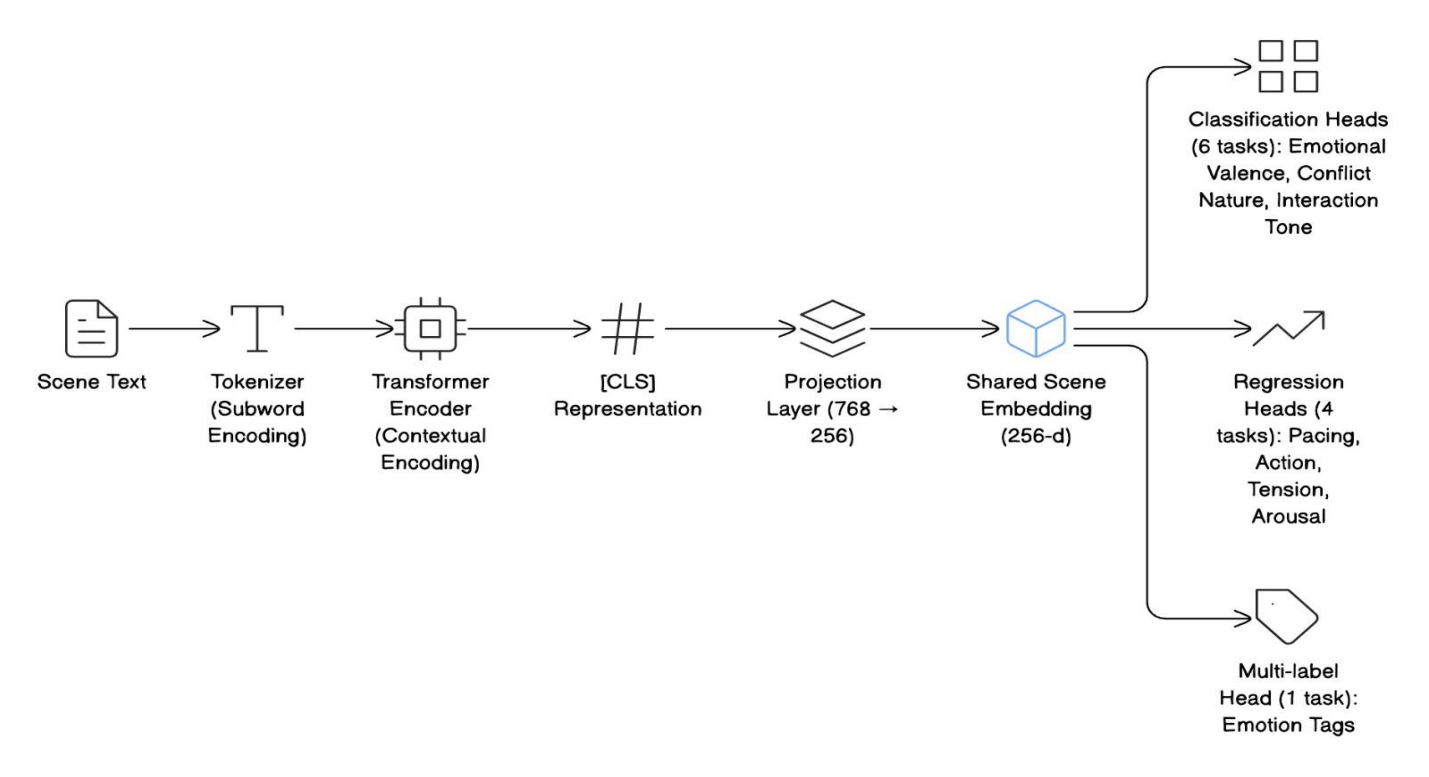
Module 1 – Scene Perception



Objective

- Extract rich scene-level representation from raw script
- Capture emotion, context, and intensity

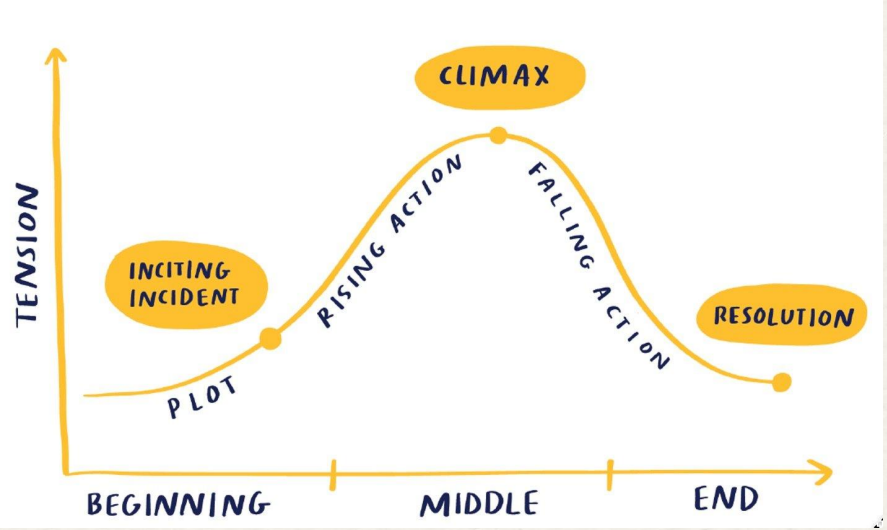
Architecture



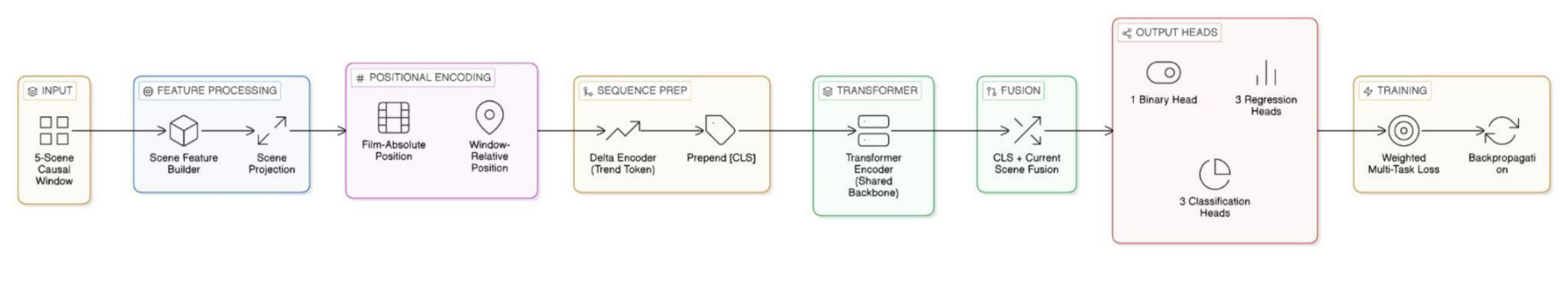
Module 2 – Narrative Modeling

Objective

- Capture narrative progression across scenes
- Model how emotion and tension evolve over time



Architecture

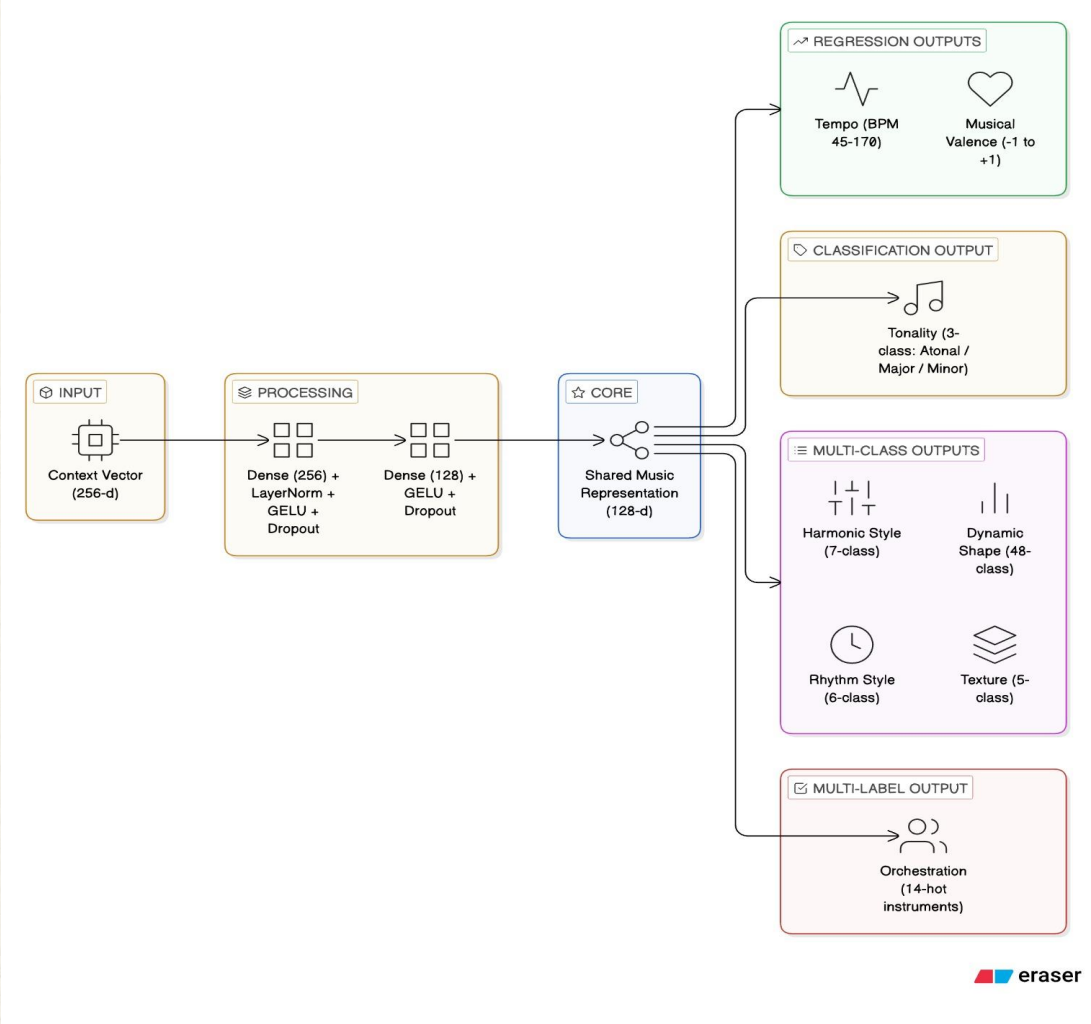


Module 3 - Music Generation Engine

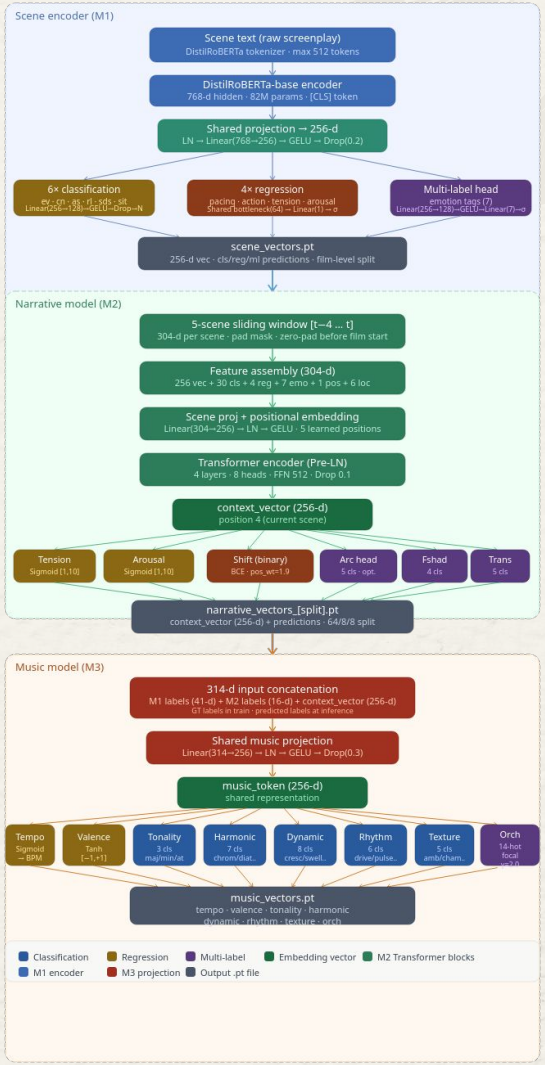
Objective

- Map narrative understanding → music characteristics
- Generate structured background music descriptors

Architecture



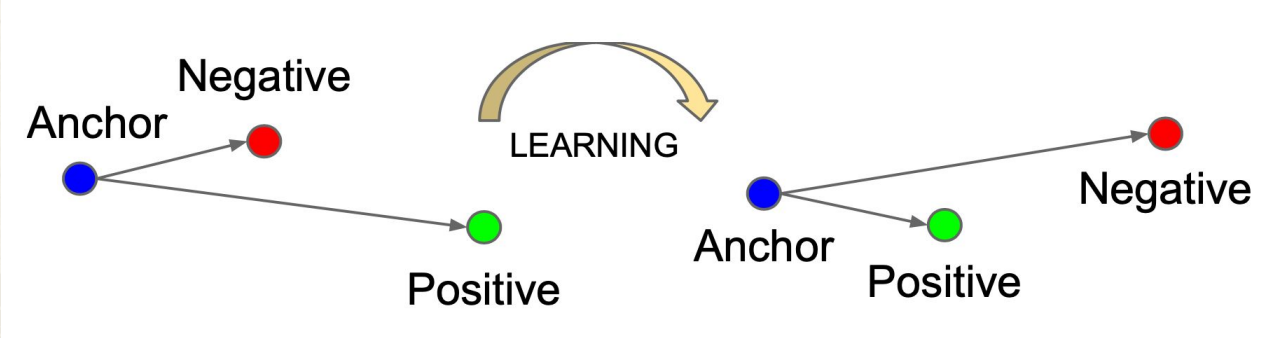
Flagship Model Architecture



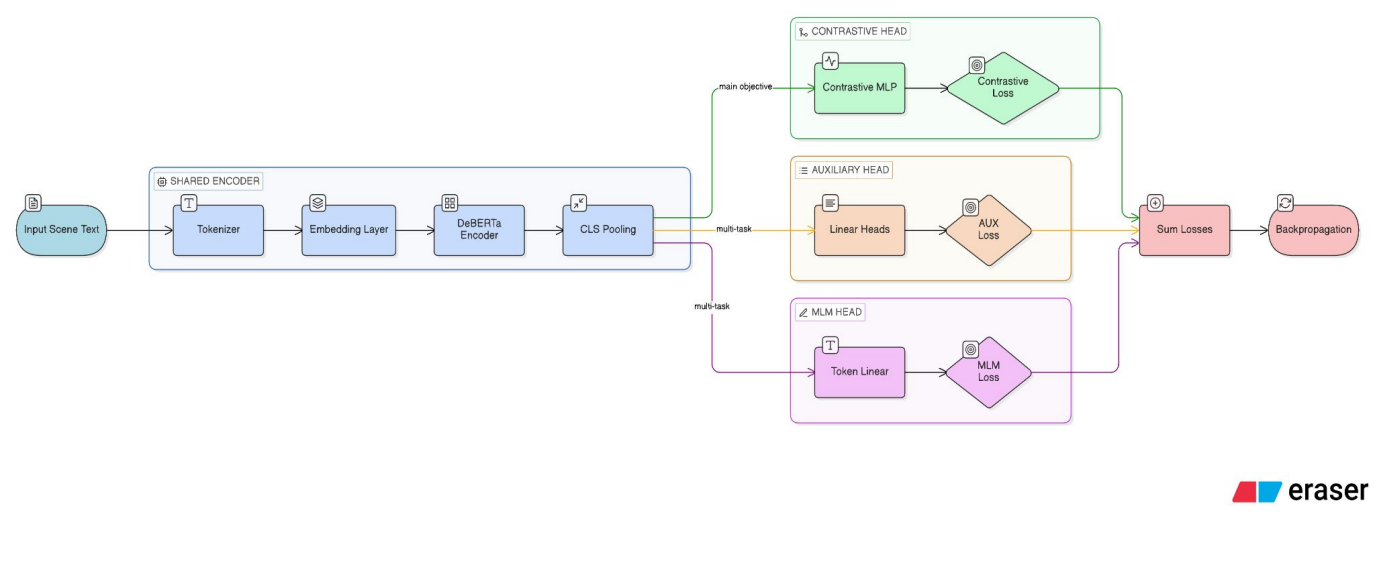
Module 0 – Representation Learning (Contrastive Learning)

Objective

- Learn meaningful scene embeddings before task-specific training
- Capture semantic similarity between scenes



Architecture and Methodology



What it does

- Learns semantic similarity between scenes
- Brings similar scenes closer in embedding space
- Pushes different scenes apart

Music Descriptor Mapping

Tempo (BPM: 45–170)

- Module 1: Determined by pacing and action intensity
- Module 2: Modified by arousal and tension progression
- Effect: Controls overall speed of the music

Musical Valence (–1 to +1)

- Module 1: Derived from emotional valence and emotion tags
- Module 2: Adjusted based on narrative progression
- Effect: Defines positive vs negative mood of music

Tonality (Atonal / Major / Minor)

- Module 1: Based on emotional valence
- Module 2: Shifted under high tension or narrative changes
- Effect: Controls harmonic mood (stable vs dissonant)

Dynamic Shape (8 classes)

- Module 1: Not directly determined
- Module 2: Strongly influenced by narrative arc position
- Effect: Controls intensity patterns (crescendo, decay, etc.)

Rhythm Style (6 classes)

- Module 1: Driven by action and pacing
- Module 2: Adjusted based on contextual flow
- Effect: Determines rhythmic energy and density

Texture (5 classes)

- Module 1: Based on acoustic space and reality layer
- Module 2: Modified by scene evolution and layering
- Effect: Controls density and atmosphere of sound

Orchestration (14-hot)

- Module 1: Determined by emotion tags and scene context
- Module 2: Refined through narrative buildup and transitions
- Effect: Controls instrument selection and layering

Witnesses

- https://drive.google.com/file/d/1hTlK8-4bwXYF98DfxWLHvIFncrxuXgvk/view
- https://drive.google.com/drive/folders/1lFkWrf7ETe7JE-fF7TvibpYCfwXks6uK?usp=sharing